Entanglement of formation for two isotropic factors

Abstract

Entanglement of formation is additive for a product of two isotropic states, with independently chosen dimensions at least two and arbitrary fidelities. A two-factor fidelity bound for unrestricted complex pure states passes to the convex roof and yields the equality. The result treats two factors; it does not assert additivity for arbitrary tensor powers.

Terhal and Vollbrecht determined the entanglement of formation of a single isotropic state [1]. The theorem here proves additivity for a product of two isotropic factors, allowing different dimensions and fidelities. An earlier additivity claim by Terhal was withdrawn because its argument for Theorem 1 was insufficient [3]; we do not use that claim. The present proof treats two factors only. It does not establish additivity for arbitrary tensor powers or identify the regularized entanglement cost.

1 Definitions and theorem

All spaces are finite-dimensional complex Hilbert spaces. Traces and Schatten norms are unnormalized. All entropies in this paper use natural logarithms, with $0 \log 0 = 0$. On $\mathbb{C}^d \otimes \mathbb{C}^d$, $d \geq 2$, set

$$|\phi_d\rangle = \frac{1}{\sqrt{d}} \sum_{a=1}^{d} |a, a\rangle, \qquad \Phi_d = |\phi_d\rangle\langle\phi_d|, \qquad \sigma_{d,F} = F\Phi_d + \frac{1-F}{d^2-1}(I - \Phi_d), \quad 0 \leq F \leq 1. \tag{1.1}$$

This is the usual isotropic family [1]. Its parameter is the actual Bell fidelity: $\operatorname{Tr} \sigma_{d,F}\Phi_d = F$. For a bipartite density ρ, define the unrestricted finite pure-state convex roof

$$E_F(\rho) = \inf_{\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|} \sum_j p_j S(\operatorname{Tr}_B |\psi_j\rangle\langle\psi_j|), \qquad S(\tau) = -\operatorname{Tr} \tau \log \tau. \tag{1.2}$$

The probabilities are nonnegative and sum to one, every ψ_j is a unit complex vector, and the barycenter equality is the actual density equality. Set $G_d(F) = E_F(\sigma_{d,F})$.

For two factors of independent dimensions $d_1, d_2 \geq 2$, regroup the physical registers as $A_1A_2 : B_1B_2$. Every normalized pure vector has a coefficient matrix M on $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$:

$$|\psi_M\rangle = \sum_{a,b,c,e} M_{ab,ce} |a, b\rangle_A |c, e\rangle_B, \qquad \|M\|_{\mathrm{HS}}^2 = \operatorname{Tr} M^* M = 1. \tag{1.3}$$

Its pure entanglement entropy is $S(MM^*) = S(M^*M)$. Partial traces $\operatorname{Tr}_1 M$ and $\operatorname{Tr}_2 M$ here are traces of the coefficient operator, not of the density $|\psi_M\rangle\langle\psi_M|$. Direct Bell contraction gives

$$f_1(M) = \langle\psi_M|(\Phi_{d_1} \otimes I)|\psi_M\rangle = \frac{\|\operatorname{Tr}_1 M\|_{\mathrm{HS}}^2}{d_1}, \qquad f_2(M) = \frac{\|\operatorname{Tr}_2 M\|_{\mathrm{HS}}^2}{d_2}. \tag{1.4}$$

Each lies in $[0, 1]$.

Theorem 1.1 (Two-factor fidelity bound)**.** *Every normalized complex matrix M in (1.3) satisfies*

$$S(M^*M) \geq G_{d_1}(f_1(M)) + G_{d_2}(f_2(M)). \tag{1.5}$$

For every mixed density ρ on these registers, with actual pair fidelities $F_i = \operatorname{Tr} \rho(\Phi_{d_i} \otimes I)$, one has

$$E_F(\rho) \geq G_{d_1}(F_1) + G_{d_2}(F_2). \tag{1.6}$$

Corollary 1.2 (Exact two-factor additivity)**.** *For every $d_1, d_2 \geq 2$ and independently chosen $F_1, F_2 \in [0,1]$,*

$$E_F(\sigma_{d_1,F_1} \otimes \sigma_{d_2,F_2}) = E_F(\sigma_{d_1,F_1}) + E_F(\sigma_{d_2,F_2}). \tag{1.7}$$

The bipartition is $A_1A_2 : B_1B_2$, and all three roofs are the unrestricted complex roofs in (1.2).

2 Finite one-factor preparations

For a probability vector $q = (q_1, \ldots, q_d)$, define

$$H(q) = -\sum_a q_a \log q_a, \qquad t_d(q) = \frac{1}{d}\left(\sum_a \sqrt{q_a}\right)^2, \qquad M_q = \operatorname{diag}(\sqrt{q_a}). \tag{2.1}$$

The pure vector of M_q has entropy $H(q)$ and Bell fidelity $t_d(q)$. Cauchy–Schwarz gives $0 \leq t_d(q) \leq 1$, with equality $t_d(q) = 1$ if and only if q is uniform.

Lemma 2.1 (Actual finite preparations)**.** *For every probability vector q,*

$$G_d(t_d(q)) \leq H(q). \tag{2.2}$$

Also $G_d(0) = 0$, $0 \leq G_d(F) \leq \log d$, and G_d is convex on $[0,1]$.

Proof. We construct a finite ensemble of local-unitary images with the stated barycenter. Write $D = \sum_a |a,a\rangle\langle a,a|$, the diagonal-pair projector, and define

$$R(F,Q) = F\Phi_d + \frac{Q}{d-1}(D - \Phi_d) + \frac{1-F-Q}{d(d-1)}(I - D). \tag{2.3}$$

Average $|\psi_{M_q}\rangle\langle\psi_{M_q}|$ over the $d!$ simultaneous physical permutations $P \otimes \overline{P}$. The diagonal-pair diagonal entries become $1/d$. Its distinct diagonal-pair entries become

$$\frac{(\sum_a \sqrt{q_a})^2 - 1}{d(d-1)}.$$

All other entries vanish. Hence this finite average is $R(t_d(q), 1 - t_d(q))$.

Let U be the normalized Fourier matrix, $U_{ab} = d^{-1/2}\exp(2\pi iab/d)$, with indices modulo d. It is unitary and $|U_{ab}|^2 = 1/d$. Let $\mathcal{T}$ be the average over the 4^d independent fourth-root diagonal phases L, acting physically as $L \otimes \overline{L}$. This average retains just the ordinary diagonal entries and the diagonal-pair block. In particular,

$$\mathcal{T}\big((U \otimes \overline{U})D(U \otimes \overline{U})^*\big) = \Phi_d + \frac{I - D}{d}. \tag{2.4}$$

To check the formula, the $(a,a;c,c)$ entry is $\sum_j |U_{aj}|^2|U_{cj}|^2 = 1/d$; the $(a,b;a,b)$ entry for $a \neq b$ is also $1/d$. Every other entry averages to zero because its independent phase character is nonconstant. Both I and Φ_d are fixed by these operations. Substituting (2.4) into (2.3) therefore gives

$$\mathcal{T}\big((U \otimes \overline{U})R(F,Q)(U \otimes \overline{U})^*\big) = R\left(F, \frac{1-F-Q}{d}\right). \tag{2.5}$$

Mix the first ensemble with weight $1/(d+1)$ and the Fourier–phase ensemble with weight $d/(d+1)$. The resulting Q coefficient is

$$\frac{Q}{d+1} + \frac{d}{d+1}\frac{1-F-Q}{d} = \frac{1-F}{d+1}.$$

Both complementary coefficients of (2.3) are then $(1-F)/(d^2-1)$, so the barycenter is exactly $\sigma_{d,F}$. All ensemble members are local-unitary images of the original pure vector, with its normalization and entropy unchanged. This proves (2.2) by the finite convex-roof definition.

For $G_d(0)$ start instead with the product vector $|a,b\rangle$, $a \neq b$. Its permutation average is $(I-D)/(d(d-1)) = R(0,0)$. The same Fourier–phase construction and mixture give $\sigma_{d,0}$ as an ensemble of product vectors of zero entropy. Thus $G_d(0) = 0$.

Every density has a finite spectral pure decomposition, each pure entropy lies in $[0, \log d]$, and concatenating finite near-optimal ensembles proves convexity of E_F. Since $\sigma_{d,F}$ is affine in F, these observations give the remaining assertions about G_d. □

3 Scalar entropy supports including the endpoints

Lemma 3.1 (Universal approximate scalar supports)**.** *For every $F \in [0,1]$ and $\epsilon > 0$, there are real a, b with $b \geq 0$ such that*

$$a + bF \geq G_d(F) - \epsilon, \tag{3.1}$$

and, for every probability vector q of length d,

$$a + bt_d(q) \leq H(q). \tag{3.2}$$

Proof. For $0 < F < 1$, a finite convex function has a supporting line on its interval: choose $a + bF = G_d(F)$ and $a + bx \leq G_d(x)$ for every $x \in [0,1]$. This elementary convex fact follows by choosing a slope between the supremum of left secant slopes and the infimum of right secant slopes. Evaluation at zero gives $a \leq G_d(0) = 0$. Since $G_d(F) \geq 0$, the identity $bF = G_d(F) - a$ gives $b \geq 0$. Lemma 2.1 then proves (3.2). For $F = 0$, the zero line has both required properties.

For $F = 1$, put $c = \log d - \epsilon$. If $c \leq 0$, the zero line works because $G_d(1) \leq \log d$. If $c > 0$, consider the compact set

$$B_c = \{q : q_a \geq 0,\ \sum_a q_a = 1,\ H(q) \leq c\}.$$

It contains probability vertices. Continuity of t_d and H gives $r = \max_{q \in B_c} t_d(q)$. One has $r < 1$: fidelity one forces the uniform vector, whose entropy is $\log d > c$. Choose

$$b = \frac{c}{1-r}, \qquad a = c - b.$$

For $q \in B_c$, $a + bt_d(q) \leq a + br = 0 \leq H(q)$. For $q \notin B_c$, $a + bt_d(q) \leq a + b = c < H(q)$. Thus (3.2) holds, while $a + b = c \geq G_d(1) - \epsilon$. No differentiability or finite endpoint tangent is needed. □

4 The product-diagonal entropy estimate

Fix two scalar supports $\ell_i(x) = a_i + b_i x$, $b_i \geq 0$, satisfying (3.2) in dimensions d_i. We will prove the affine estimate

$$S(M^*M) \geq a_1 + a_2 + b_1 f_1(M) + b_2 f_2(M) \tag{4.1}$$

for every normalized complex M.

Lemma 4.1 (Product diagonal)**.** *The estimate* (4.1) *holds for* $M = \operatorname{diag}(\sqrt{p_{xy}})$*, where* p *is any probability distribution on* $\{1, \dots, d_1\} \times \{1, \dots, d_2\}$*.*

Proof. Its entropy is $H(X, Y)$. Let $p_y = \sum_x p_{xy}$. For every $p_y > 0$, apply the first scalar support to the conditional distribution $p_{x|y} = p_{xy}/p_y$:

$$H(X \mid Y = y) \geq a_1 + \frac{b_1}{d_1} \left(\sum_x \sqrt{p_{x|y}} \right)^2 .$$

After weighting by p_y and summing,

$$H(X \mid Y) \geq a_1 + b_1 f_1(M), \qquad H(Y \mid X) \geq a_2 + b_2 f_2(M). \tag{4.2}$$

Zero-weight conditionals are omitted; their entire column or row is zero. The fidelity identities are

$$f_1(M) = \frac{1}{d_1} \sum_y \left(\sum_x \sqrt{p_{xy}} \right)^2 , \qquad f_2(M) = \frac{1}{d_2} \sum_x \left(\sum_y \sqrt{p_{xy}} \right)^2 .$$

Finally $H(X, Y) - H(X \mid Y) - H(Y \mid X) = I(X : Y) \geq 0$. Adding (4.2) proves the result. This conditional entropy argument is the product-basis argument represented in [2, Section III, Eqs. (18)–(32)]. □

5 Weighted spectral comparison for positive coefficients

Lemma 5.1 (Two-site spectral comparator)**.** *Let* $P \geq 0$ *be a coefficient matrix on* $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$*, and let* $w_1, w_2 \geq 0$*. There is a positive matrix* P' *diagonal in a product basis, with the same eigenvalues as* P*, such that*

$$w_1 \| \operatorname{Tr}_1 P \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 P \|_{\mathrm{HS}}^2 \leq w_1 \| \operatorname{Tr}_1 P' \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 P' \|_{\mathrm{HS}}^2 . \tag{5.1}$$

Proof. Put $A = \operatorname{Tr}_1 P$, $B = \operatorname{Tr}_2 P$, and define the Hermitian site field

$$K = w_1 I_1 \otimes A + w_2 B \otimes I_2 .$$

The two summands act on separate factors. Diagonalize A and B in their own orthonormal eigenbases. This makes K diagonal in a product basis, with entries $h_{xy} = w_1 \mu_y + w_2 \nu_x$. If λ_j are the eigenvalues of P, its diagonal entries in this basis are $\sum_j |U_{ij}|^2 \lambda_j$, for a unitary U. The matrix $(|U_{ij}|^2)$ is doubly stochastic. A linear functional on doubly stochastic matrices attains its maximum at a permutation matrix, by the Birkhoff decomposition. Hence some assignment of the λ_j to the product basis gives P' with

$$\operatorname{Tr} P' K \geq \operatorname{Tr} P K = w_1 \|A\|_{\mathrm{HS}}^2 + w_2 \|B\|_{\mathrm{HS}}^2 =: C(P). \tag{5.2}$$

For clarity, this assignment depends on the given matrix and the chosen weights; it retains its complete spectrum.

Define $C(Q) = w_1 \| \operatorname{Tr}_1 Q \|_{\mathrm{HS}}^2 + w_2 \| \operatorname{Tr}_2 Q \|_{\mathrm{HS}}^2$. Completing the two independent Hilbert–Schmidt squares gives

$$\begin{aligned} C(P') &\geq 2 w_1 \operatorname{Tr}(\operatorname{Tr}_1 P') A + 2 w_2 \operatorname{Tr}(\operatorname{Tr}_2 P') B - C(P) \\ &= 2 \operatorname{Tr} P' K - C(P) \geq C(P). \end{aligned}$$

This proves (5.1). Vanishing weights and zero eigenvalues are allowed throughout. □

Lemma 5.2 (Positive affine estimate)**.** *The estimate* (4.1) *holds for every* $M = P \geq 0$ *with* $\|P\|_{\mathrm{HS}} = 1$.

Proof. Apply Lemma 5.1 with $w_i = b_i/d_i$. The resulting P' has the same squared spectrum and therefore $S(P'^2) = S(P^2)$ and $\|P'\|_{\mathrm{HS}} = 1$, while

$$b_1 f_1(P) + b_2 f_2(P) \leq b_1 f_1(P') + b_2 f_2(P').$$

Its nonnegative diagonal eigenvalues can be written $\sqrt{p_{xy}}$ with $\sum p_{xy} = 1$. Lemma 4.1 applies in that product basis. Product coefficient conjugations preserve both partial-trace norms and the squared spectrum; thus its estimate applies to P' in the original coordinates as well. Combining the inequalities proves (4.1) for P. □

6 Both polar moduli and unrestricted complex coefficients

Lemma 6.1 (Partial-trace modulus bound)**.** *For every complex square coefficient matrix* M*, write* $L = \sqrt{MM^*}$ *and* $R = \sqrt{M^*M}$*. Then*

$$f_i(M) \leq \frac{f_i(L) + f_i(R)}{2}, \qquad i = 1, 2. \tag{6.1}$$

If $\|M\|_{\mathrm{HS}} = 1$*, both moduli have Hilbert–Schmidt norm one and their pure entropies equal* $S(M^*M)$.

Proof. The literal block matrix

$$\begin{pmatrix} L & M \\ M^* & R \end{pmatrix} \geq 0 \tag{6.2}$$

is positive. For example, a singular-value decomposition $M = U\Sigma V^*$ writes it as $\binom{U}{V}\Sigma \begin{pmatrix} U^* & V^* \end{pmatrix}$. Applying the completely positive partial trace to its blocks gives

$$\begin{pmatrix} A_i & X_i \\ X_i^* & B_i \end{pmatrix} \geq 0, \qquad A_i = \mathrm{Tr}_i\, L, \quad B_i = \mathrm{Tr}_i\, R, \quad X_i = \mathrm{Tr}_i\, M.$$

A positive block matrix admits $X_i = A_i^{1/2} T_i B_i^{1/2}$ with $\|T_i\|_\infty \leq 1$. This includes singular diagonal blocks: restrict to their supports, or obtain the statement by adding δI and taking a convergent subsequence of contractions as $\delta \downarrow 0$. Schatten Hölder gives

$$\|X_i\|_{\mathrm{HS}} \leq \|A_i^{1/2}\|_4 \|T_i\|_\infty \|B_i^{1/2}\|_4, \qquad \|X_i\|_{\mathrm{HS}}^2 \leq \|A_i\|_{\mathrm{HS}} \|B_i\|_{\mathrm{HS}}.$$

Arithmetic–geometric mean and division by d_i give (6.1). The equal norm and entropy claims follow from the common singular values, including all zero values, of M. □

Proof of the affine estimate (4.1). Apply Lemma 5.2 to L and R. Average its two inequalities. Their left sides have the same entropy $S(M^*M)$. Since $b_1, b_2 \geq 0$, Lemma 6.1 gives

$$\begin{aligned} S(M^*M) &\geq a_1 + a_2 + \frac{b_1}{2}\big(f_1(L) + f_1(R)\big) + \frac{b_2}{2}\big(f_2(L) + f_2(R)\big) \\ &\geq a_1 + a_2 + b_1 f_1(M) + b_2 f_2(M). \end{aligned}$$

Thus the estimate holds for every complex coefficient, without a positivity, normality, rank, or product-basis hypothesis on M. □

7 Pure and mixed roofs

Proof of Theorem 1.1. Choose supports from Lemma 3.1 independently at $f_1(M)$ and $f_2(M)$, each within ϵ of the corresponding one-factor value. The affine estimate yields

$$S(M^*M) \geq G_{d_1}(f_1(M)) + G_{d_2}(f_2(M)) - 2\epsilon.$$

Letting $\epsilon \downarrow 0$ proves (1.5), including all fidelity endpoints.

For the mixed statement, let $\rho = \sum_j p_j |\psi_{M_j}\rangle\langle\psi_{M_j}|$ be any finite normalized complex ensemble. Physical Bell expectations are linear in the actual density barycenter:

$$\sum_j p_j f_i(M_j) = F_i. \tag{7.1}$$

Average (1.5) and use convexity of each G_{d_i}:

$$\sum_j p_j S(M_j^* M_j) \geq \sum_j p_j \big(G_{d_1}(f_1(M_j)) + G_{d_2}(f_2(M_j))\big) \geq G_{d_1}(F_1) + G_{d_2}(F_2).$$

Only the ensemble means are fixed in (7.1); individual components can have arbitrary fidelities. Taking the infimum over every such ensemble proves (1.6). □

Proof of Corollary 1.2. The pair fidelities of the actual tensor product $\sigma_{d_1,F_1} \otimes \sigma_{d_2,F_2}$ are exactly F_1, F_2, since each factor has trace one. Thus (1.6) gives the lower inequality in (1.7).

For the upper inequality choose, for each factor, a finite ensemble whose cost is within ϵ of its roof. Their product ensemble has the stated tensor density as its actual barycenter. Each product pure vector has entropy equal to the sum of its two entropies across $A_1A_2 : B_1B_2$, because its Alice density is a tensor product. Therefore the product ensemble cost is at most $G_{d_1}(F_1) + G_{d_2}(F_2) + 2\epsilon$. Taking the infimum and letting $\epsilon \downarrow 0$ proves the upper inequality. No minimizing ensemble is needed. □

References

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